Agricultural Commodity Price Analysis Dashboard (26-01-2026)

Project Overview and Objective

The project analyzes the current daily price of agricultural commodities across markets (mandis) in India using data sourced from the Government of India's open data platform. The data set provides daily observations of minimum, maximum, and modal prices for various commodities in different markets. These prices are collected through official reporting and represent the wholesale price scenario on a specific date (e.g., 26 January 2026)

The main objective of this project is to build an analytical dashboard in Power BI that enables:

Comparative Price Analysis – To compare modal, minimum, and maximum prices across commodities, markets, and states on a specific date.

Market Insights – To identify top-performing markets based on price levels.

Commodity Price Profiling – To determine which commodities command higher wholesale prices and which are lower.

Regional Price Variability – To visualize and understand how commodity prices vary regionally across different states and mandi markets.

Decision Support – To provide data-backed insights that can help traders, analysts, and policymakers make informed pricing and sourcing decisions.

Data Sources

Source Description and Timeline : https://www.data.gov.in/resource/current-daily-price-various-commodities-various-markets-mandi

Problem Statement

- Inconsistencies such as duplicates, blanks, and irrelevant fields.
- Lack of structured schema, making direct analysis difficult.
- Complexity in comparing prices across multiple dimensions (state, market, commodity) without aggregation logic.
- Need for business-friendly insights that go beyond raw figures.

Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Arrival Date	Date	Date on which commodity price was recorded

State	Text	Name of the state where maket is located
District	Text	District of the market
Market	Text	Name of the mandi/market
Commodity	Text	Name of agricultural commodity
Min_Price	Fixed Decimal Number	Lowest price recorded for the commodity in that market
Max_Price	Fixed Decimal Number	Highest price recorded for the commodity in that market
Modal_Price	Fixed Decimal Number	Most frequently occurring price in the market.
Price_Category	Text	Classification of price into High/medium/low

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Data Pre-Processing Power Query)

Data Import

The commodity price dataset was imported into Microsoft Power BI using the Get Data option from the Excel/CSV source. This enabled structured loading of raw mandi price data into the Power BI environment for further transformation and modeling.

Removal of Unnecessary Columns

Several columns were identified as non-essential for analysis.

Only relevant analytical attributes were retained:

- Date
- State
- Market
- Commodity
- Minimum Price
- Maximum Price
- Modal Price

Removed unnecessary columns:

- Reduced data model size
- Improved performance
- Increased clarity in reporting

Handling Missing Values

- Blank and null values were identified in key columns.
- Rows containing missing values in important fields (Date, State, Market, Commodity, and Price) were removed using filtering techniques in Power Query Editor by excluding blank and null values to ensure data completeness and analytical accuracy.

Text Cleaning (Data Standardization)

Text columns such as State - District - Market - Commodity were cleaned using:

TRIM – Removed extra leading and trailing spaces

CLEAN – Removed hidden or non-printable characters

Removing Duplicate Records(Dimension Tables)

Duplicate values were removed from dimension tables such as State, Market, Commodity, and Date.

Trim and Clean operations were applied to text columns (State, District, Market, Commodity).

Data Type Correction

Correct data types were assigned to ensure accurate calculations and aggregations.

Price Columns:

Minimum Price

Maximum Price

Modal Price

Converted to: **Fixed Decimal Number**

✔ Prevents rounding errors
✔ Maintains financial precision
✔ Improves calculation accuracy

Date Column:

Converted to: **Date Data Type**

✔ Enables time intelligence functions
✔ Supports filtering and chronological analysis

Calculated Column

Price Category as a calculated column in the Fact table because it is derived from row-level modal price values. Since it depends on transaction-level data, it belongs in the fact table rather than a dimension table.

Price Category =
IF(Fact_Commodity_Prices[Modal_x0020_Price]>5000,"High",IF(Fact_Commodity_Prices[Modal_x0020_Price]>2000,"Medium","Low"))

Price Spread created a Price Spread column to analyze intra-market price volatility and a Price Category column to segment commodities based on modal price levels."

Price Spread = Fact_Commodity_Prices[Max_x0020_Price]-Fact_Commodity_Prices[Min_x0020_Price]

Creation of Dimension Tables

Date Dimension (Dim_Date)

A reference of the Fact Table was used to create the Dim_Date table.

A continuous date range was generated to cover the entire analysis period. Duplicate dates were eliminated. Date data type was correctly assigned.

Since the data set represents a single-day snapshot for 2026, time-intelligence measures such as YTD and Previous Year comparison were not applicable. The analysis focuses on cross-sectional price distribution rather than trend analysis."

Date	Year	Month_No	Month_Name	Quarter	Day
01 July 2026	2026	7	July	3	1
02 July 2026	2026	7	July	3	2
03 July 2026	2026	7	July	3	3
04 July 2026	2026	7	July	3	4
05 July 2026	2026	7	July	3	5
06 July 2026	2026	7	July	3	6
07 July 2026	2026	7	July	3	7
08 July 2026	2026	7	July	3	8
09 July 2026	2026	7	July	3	9
10 July 2026	2026	7	July	3	10
11 July 2026	2026	7	July	3	11
12 July 2026	2026	7	July	3	12
13 July 2026	2026	7	July	3	13
14 July 2026	2026	7	July	3	14
15 July 2026	2026	7	July	3	15
16 July 2026	2026	7	July	3	16
17 July 2026	2026	7	July	3	17
18 July 2026	2026	7	July	3	18
19 July 2026	2026	7	July	3	19
20 July 2026	2026	7	July	3	20
21 July 2026	2026	7	July	3	21
22 July 2026	2026	7	July	3	22
23 July 2026	2026	7	July	3	23
24 July 2026	2026	7	July	3	24
25 July 2026	2026	7	July	3	25
26 July 2026	2026	7	July	3	26
27 July 2026	2026	7	July	3	27

State Dimension (Dim_State)

A separate reference of the Fact Table was used to create the Dim_Date table.

Duplicate state values were removed. Trim and Clean operations were applied to remove extra spaces and hidden characters.

A unique state (primary key) was generated to uniquely identify each state.

State
Odisha
Telangana
Uttar Pradesh
Punjab
Kerala
Haryana

Market Dimension (Dim_Market)

A separate reference of the Fact Table was used to create the Dim_Date table.

Duplicate market records were removed. Text columns were cleaned using Trim and Clean. Each market is uniquely associated with one district and one state.

Enables market-wise and district-wise price comparison.

State	District	Market
Odisha	Mayurbhanja	Karanjia APMC
Telangana	Nalgonda	Neredcherla APMC
Uttar Pradesh	Sitapur	Hargaon (Laharpur) APMC
Punjab	Patiala	Patiala APMC
Kerala	Kollam	Chathanoor APMC
Haryana	Gurgaon	Gurgaon APMC

Commodity Dimension(Dim_Commodity)

A reference of the Fact Table was used to create the Dim_Date table.

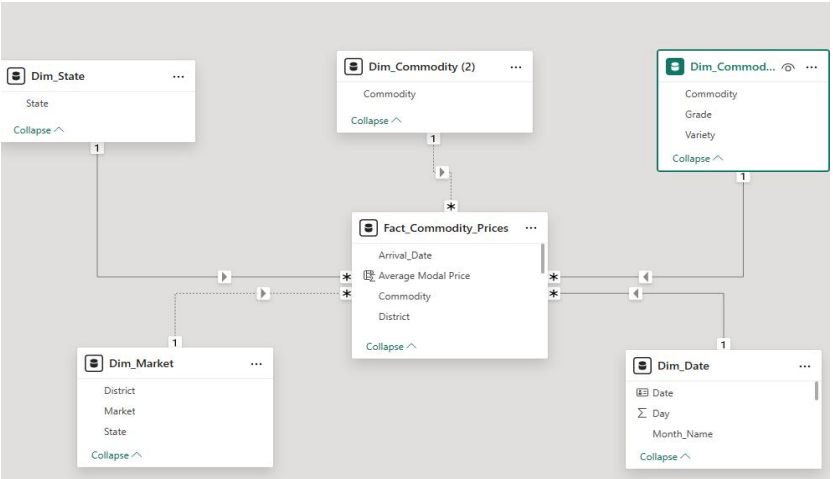
Duplicate commodity values were removed. Text formatting was standardized using Trim and Clean.

Commodity	Variety	Grade
Pumpkin	Pumpkin	FAQ
Paddy(Common)	B P T	Local
Peas Wet	Other	Grade B
Onion	Onion	Medium
Cowpea(Veg)	Cowpea (Veg)	Large
Cauliflower	Cauliflower	Grade A
Green Chilli	Green Chilly	Grade Range-2

The dataset was cleaned, standardized, and structured using proper data transformation techniques including null handling, duplicate removal, text cleaning, and data type correction. A calculated Price Category column was created at the transaction level to enable price segmentation analysis, ensuring an optimized and analytically data model

Final Data Model Preparation

- A star schema data model was created by connecting the Fact Table with Dim_Date, Dim_State, Dim_Market and Dim_Commodity using one-to-many relationships.
- The Fact Table contains Commodity prices while the dimension tables store descriptive information such as date, state and Commodity.
- This data model improves data organization, enables efficient filtering, and supports accurate analysis and visualization in Power BI dashboards.
- Below shows the relationship diagram as fact table in the centre connecting to all dimension table.



DAX Measures

Average Minimum Price

Average Minimum Price = AVERAGE(Fact_Commodity_Prices[Min_x0020_Price])

Purpose: Calculates the average of daily minimum prices across markets to understand the lower price level trend.

Average Maximum Price

Average_Maximum Price = AVERAGE(Fact_Commodity_Prices[Max_x0020_Price])

Purpose: Computes the average of daily maximum prices to analyze the upper price range across markets.

Total Modal Price

Total_Modal Price = SUM(Fact_Commodity_Prices[Modal_x0020_Price])

Purpose: Aggregates modal prices to measure total price contribution across selected filters.

Avg Modal Price

Avg Modal Price = average(Fact_Commodity_Prices[Modal_x0020_Price])

Purpose: Determines the average modal price to compare price levels across commodities, markets, or states.

Distinct Market Count

Distinct_Market Count = DISTINCTCOUNT(Fact_Commodity_Prices[Market])

Purpose: Counts the number of unique markets involved in the dataset.

Market Count

Market_Count = COUNT(Fact_Commodity_Prices[Market])

Purpose: Counts total market records, including repeated entries.

Maximum Price

Maximun_Price = MAX(Fact_Commodity_Prices[Modal_x0020_Price])

Purpose: Identifies the highest modal price recorded in the dataset.

Minimum Price

Minimum_Price = min(Fact_Commodity_Prices[Modal_x0020_Price])

Purpose: Identifies the lowest modal price recorded in the dataset

Price Range (Correction Needed)

Price_Range = [Maximun_Price] - [Minimum_Price]

Purpose: Calculates the difference between highest and lowest modal price to measure price volatility.

Market Rank

Market Rank = RANKX(ALL(Dim_Market[Market]),[Total_Modal Price],,desc)

Purpose: Ranks markets based on total modal price in descending order.

Top 5 Market Price

Top 5 Market Price = if([Market Rank]<=5,[Total_Modal Price],BLANK())

Purpose: Displays total modal price only for the top 5 ranked markets.

Analysis and Visualizations (Power BI)

KPI Overview

A total of 5 KPI includes using card visuals

The **Average Modal Price** stands at ₹3.15K, indicating the general price benchmark across commodities.

The **Maximum recorded** price is ₹15K, showing presence of premium-priced commodities in select markets.

The **Minimum price** is ₹200, highlighting wide pricing disparity across products.

The **Price Range** of ₹14.8K reflects significant intra-day price variation.

17 distinct markets contribute to the overall pricing landscape, showing moderate geographic coverage.

Insight: The large price spread suggests strong variation between high-value and low-value agricultural commodities.

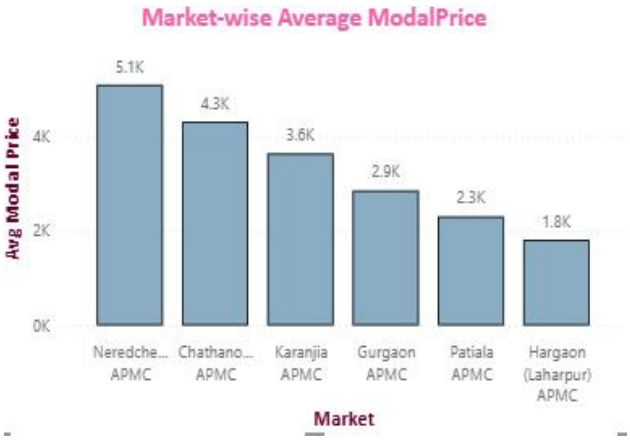
CHART ANALYSIS

Commodity Wise Average Modal_Price

Insights ·

- Green Chilli commands the highest average modal price, indicating strong consumer demand and premium market positioning.
- Pumpkin records the lowest average price, suggesting higher supply availability or lower demand intensity.
- Significant price gap between commodities reflects profitability differences, influencing farmers' crop selection decisions.
- High-value commodities like Green Chilli present better revenue opportunities for traders and producers.

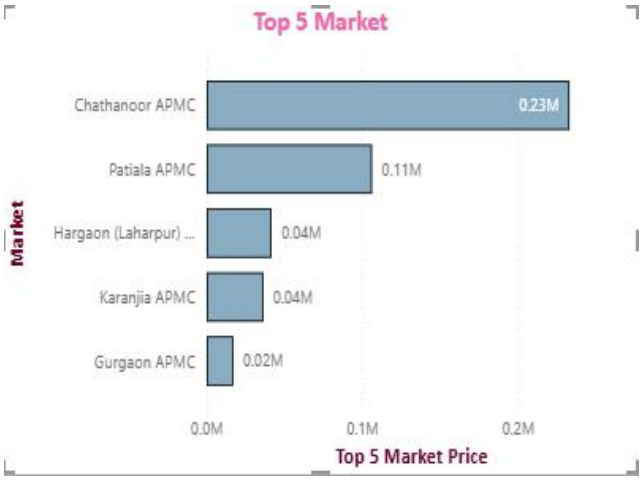
Market-wise Average Modal Price



Insights

- Reveals price variation across different markets.
- Markets with higher averages may indicate strong demand or limited supply.
- Lower-priced markets may indicate oversupply or weaker buyer demand.

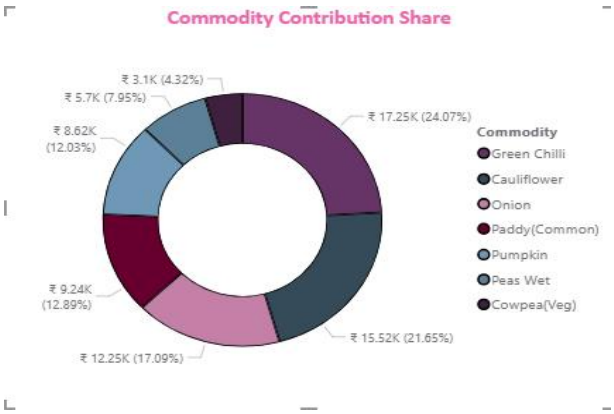
Top 5 Markets by Modal Price



Insights

- Certain markets consistently show higher average prices, reflecting stronger purchasing power and better trade activity.
- Lower-priced markets may indicate oversupply or weaker buyer demand

Commodity Contribution Share (Donut

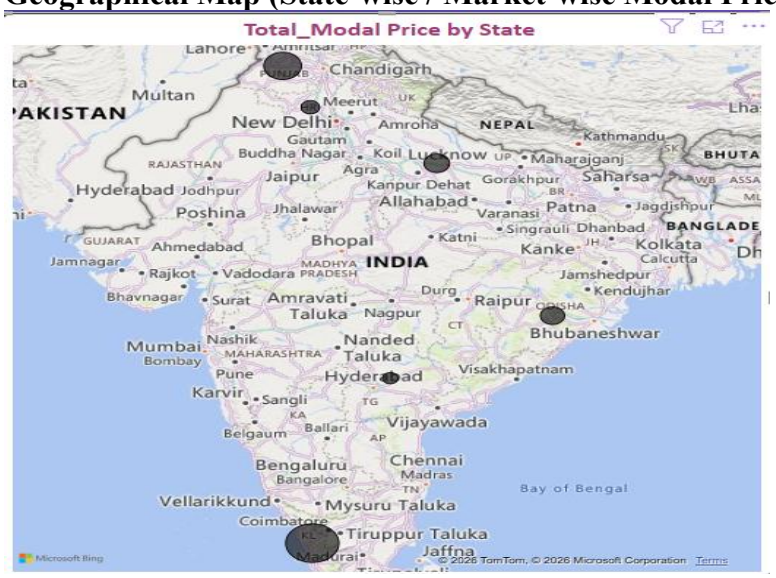


Insights

- Green Chilli contributes the largest share of total modal price, making it the dominant commodity.
- Smaller segments represent low-impact commodities.
- Price contribution is concentrated among a few major commodities.
- Commodity distribution is uneven, indicating varying economic impact.

The 4 Commodity wise analysis highlights substantial price disparities across commodities and markets. Green Chilli emerges as a premium-priced commodity, while Chatanoor APMC leads among markets.

Geographical Map (State-wise / Market-wise Modal Price



Insights

- Visualizes regional distribution of commodity prices across India.
- Larger bubbles indicate states/markets with higher modal prices.
- Kerala shows the highest modal price concentration, emerging as a premium agricultural pricing zone.
- Price distribution across other states remains comparatively moderate, indicating uneven revenue concentration.

State-wise Modal Price Comparison

Insights

- Telangana records the highest modal price, indicating strong demand conditions and premium market valuation.
- Uttar Pradesh shows the lowest modal price, suggesting relatively higher supply or lower competitive pricing.
- States with higher modal prices offer better revenue realization for farmers and traders.

Price Distribution Table (With Conditional Formatting

Commodity	Market	Average Modal Price	Maximun_Price	Minimum_Price	Price_Range
	Chathanoor APMC	$3,153.56	₹ 15,000	₹ 1,600	
Cauliflower	Chathanoor APMC	$3,153.56	₹ 3,200	₹ 3,200	₹ 0
Cowpea(Veg)	Chathanoor APMC	$3,153.56	₹ 3,100	₹ 3,100	₹ 0
Green Chilli	Chathanoor APMC	$3,153.56	₹ 5,500	₹ 5,500	₹ 0
Onion	Chathanoor APMC	$3,153.56	₹ 3,500	₹ 2,600	₹ 900
Pumpkin	Chathanoor APMC	$3,153.56	₹ 4,000	₹ 1,900	₹ 2,100
	Gurgaon APMC	$3,153.56	₹ 7,500	₹ 650	6,850
Onion	Gurgaon APMC	$3,153.56	₹ 1,500	₹ 1,500	₹ 0
	Hargaon (Laharpur) APMC	$3,153.56	₹ 7,300	₹ 310	6,990
Cauliflower	Hargaon (Laharpur) APMC	$3,153.56	₹ 1,650	₹ 1,650	₹ 0
Green Chilli	Hargaon (Laharpur) APMC	$3,153.56	₹ 3,800	₹ 3,800	₹ 0
Onion	Hargaon (Laharpur) APMC	$3,153.56	₹ 750	₹ 750	₹ 0
Paddy(Common)	Hargaon (Laharpur) APMC	$3,153.56	₹ 2,369	₹ 2,102	₹ 267
Pumpkin	Hargaon (Laharpur) APMC	$3,153.56	₹ 900	₹ 900	₹ 0
	Karanjia APMC	$3,153.56	₹ 5,000	₹ 2,200	₹ 2,800
Cauliflower	Karanjia APMC	$3,153.56	₹ 5,200	₹ 5,200	₹ 0
Pumpkin	Karanjia APMC	$3,153.56	₹ 1,000	₹ 820	₹ 180
	Neredcherla APMC	$3,153.56	₹ 7,800	₹ 7,800	₹ 0
Paddy(Common)	Neredcherla APMC	$3,153.56	₹ 2,400	₹ 2,400	₹ 0
	Patiala APMC	$3,153.56	₹ 13,000	₹ 200	
Cauliflower	Patiala APMC	$3,153.56	₹ 2,500	₹ 400	₹ 2,100
Green Chilli	Patiala APMC	$3,153.56	₹ 4,000	₹ 1,450	₹ 2,550
Onion	Patiala APMC	$3,153.56	₹ 2,200	₹ 400	₹ 1,800
Peas Wet	Patiala APMC	$3,153.56	₹ 2,100	₹ 1,600	₹ 500
Total			**₹ 15,000**	**₹ 200**	**₹ 14,800**

Insights

- Wide variation between minimum and maximum prices highlights pricing volatility.
- Certain commodities show stable pricing with narrow price range.
- High price spread commodities carry higher revenue opportunity but also higher volatility risk.
- Applied conditional formatting data bars to visually highlight price variations.

Treemap

Insights

- Displays hierarchical price contribution visually.
- Larger blocks represent higher price contribution.
- Quickly identifies dominant commodities or markets.
- Helps compare contribution proportion within limited screen space.
- Highlights major revenue-driving categories.

Slicers & Filters Implemented

These slicers allow dynamic filtering across all visuals in the dashboard.

Commodity State Market

Commodity Slicer The Commodity slicer enables users to

- Filter the dashboard by a specific agricultural commodity.
- Compare price performance of one commodity across multiple markets and states.
- Identify high-performing or low-performing commodities quickly
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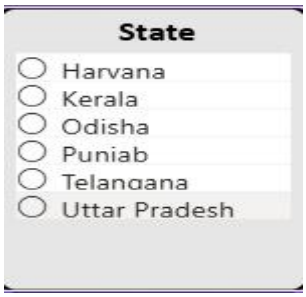
State Slicers The State slicer allows users to:

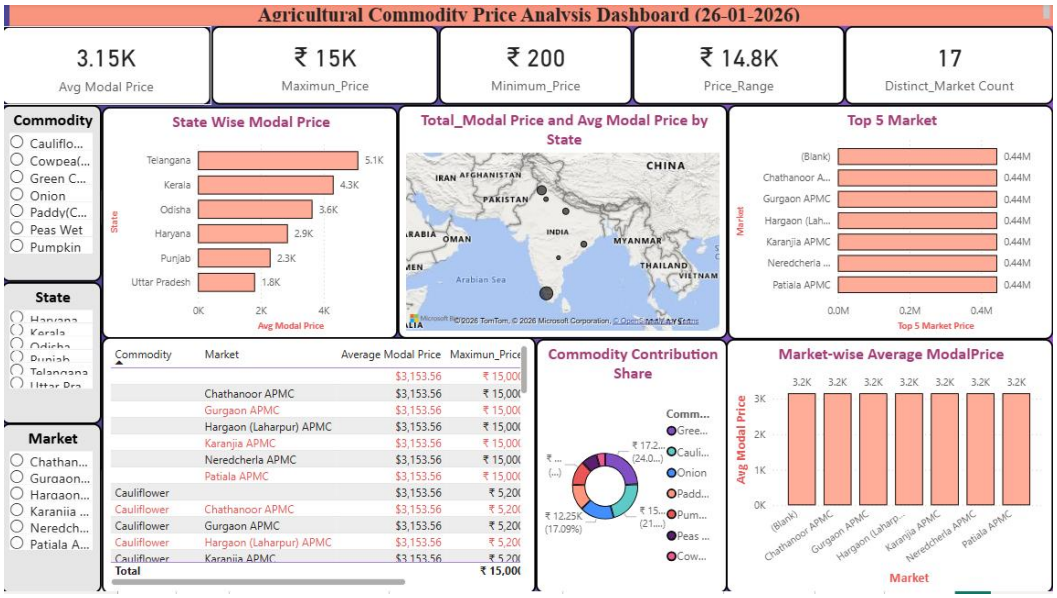
- View price performance specific to a selected state.
- Analyze state-level pricing strength and concentration.
- Compare market performance within a particular state.



Market Slicers The Market slicer enables:

- Filtering data by specific APMC markets.
- Analyzing price variation within selected markets.
- Identifying top-performing and underperforming markets.

Agricultural Commodity Price Analysis Dashboard (26-01-2026)



Insights

Revenue - Agricultural price value is concentrated in select commodities and markets, indicating dependency on limited high-performing segments

Regional - Strong pricing differences across states reflect regional demand-supply imbalance and uneven agricultural market strength.

Profitability - Farmers cultivating high-value commodities like Green Chilli may achieve significantly higher revenue compared to low-value crops like Pumpkin.

Supplying commodities to high-performing markets such as Chatanoor APMC can improve price realization and profitability.

Large price spread across commodities indicates varying risk levels; high-range commodities may offer high returns but involve greater pricing fluctuation.

Conclusion

This one-day agricultural price analysis reveals substantial variation across commodities, markets, and states. Pricing power is concentrated in select regions such as Kerala and Telangana, while a few commodities dominate overall value generation. The findings highlight the importance of strategic crop selection and market targeting to maximize revenue potential in agricultural trade.